MINI PROJECT

Chocolate Sales Performance Analysis

Project Overview

This interactive visualisation provides an overview of Chocolate sales performance. It analyses revenue, profit, sales trends, product performance, regional sales, and key business insights to support data-driven decision-making.

Prepared by: Shaheeka Palorath Kandiyil

Tools: Excel and Power BI

Dataset: https://www.kaggle.com/datasets/arjunmehta1992/chocolate-sales-in-20222023

Chocolate Sales Summary (2022-23)

Order_ID	Order_Date	Salesperson	Product	Country	Sum of Discount %	Sum of Marketing Spend	Sum of Boxes Shipped	Sum of Estimated Net Sales	Sum of Amount	Column	Country
ORD-000001	17/12/2023	Sora Watanabe	White Choco Bar	Brazil	6.80%	$76.03	99.00	$1,301.9541	$313.13	Average	Brazil
ORD-000002	02/03/2022	Yuki Sato	Mixed Assortment Box	Germany	1.70%	$43.14	190.00	$535.7089	$764.71	Average	Germany
ORD-000003	11/05/2023	Rafael Oliveira	Milk Classic Bar	Australia	3.40%	$53.83	80.00	$189.5355	$473.78	Average	Australia
ORD-000004	05/05/2023	Priya Sharma	Pralines Gift Box	Australia	7.30%	$127.85	92.00	$309.8137	$308.26	Average	Australia
ORD-000005	23/04/2022	Lucas Walker	70% Dark Bar	Germany	19.40%	$44.82	124.00	$367.5655	$1,628.45	Average	Germany
Total					2621491.40%	$19,089,674.0498	27,916,821.00	$117,983,804.7194018	$102,572,394.94		

VLOOKUP SAMPLE DATA (ORD-042100-043500)

Order_ID	Sales Person	Product	Country	Sum of Discount %	Order_Date	Sum of Marketing Spend	Sum of Boxes Shipped	Sum of Estimated Net Sales	Sum of Amount
ORD-042103	Yuki Sato	70% Dark Bar	Australia	12.53	04/10/2023	$95.48	31	$388.2047	$424.14
ORD-042122	Yuki Sato	Truffle Gift Box	Australia	2.94	08/08/2023	$37.36	126	$370.2955	$293.49
ORD-042151	Yuki Sato	70% Dark Bar	Australia	13.42	02/02/2023	$94.06	46	$617.1718	$457.47
ORD-042181	Yuki Sato	70% Dark Bar	Australia	2.75	26/03/2023	$422.8	219	$601.7983	$1,692.16
ORD-042252	Yuki Sato	Choco Coins Bag	Australia	3.39	06/09/2022	$49.88	153	$517.9127	$152.35
ORD-042262	Yuki Sato	70% Dark Bar	Germany	2.75	01/09/2023	$157.86	110	$302.3367	$252.33
ORD-042328	Yuki Sato	Hazelnut Milk Bar	India	3.10	29/09/2023	$39.83	75	$232.4116	$204.95
ORD-042337	Yuki Sato	Mixed Assortment Box	Australia	3.56	30/09/2023	$32.98	426	$1,516.1505	$404.32
ORD-042354	Yuki Sato	70% Dark Bar	Australia	2.99	19/08/2023	$84.24	91	$271.5077	$741.37
ORD-042360	Yuki Sato	Mixed Assortment Box	Germany	4.18	05/11/2023	$54.38	142	$592.9308	$514.09
ORD-042413	Yuki Sato	Mixed Assortment Box	Australia	17.51	20/03/2023	$81.41	76	$1,328.7106	$874.76
ORD-042444	Yuki Sato	Truffle Gift Box	Germany	3.88	24/06/2023	$87.37	209	$809.3387	$534.92
ORD-042487	Yuki Sato	Almond Crunch Bar	Australia	2.88	16/02/2023	$151.09	360	$1,034.7471	$973.36
ORD-042502	Yuki Sato	70% Dark Bar	Brazil	18.09	30/05/2023	$62.84	57	$1,029.6658	$287.04
ORD-042506	Yuki Sato	Milk Classic Bar	Australia	2.83	04/10/2022	$57.17	84	$237.411	$555.76
ORD-042562	Yuki Sato	70% Dark Bar	Brazil	2.65	08/04/2023	$122.78	268	$708.9287	$467.96
ORD-042583	Yuki Sato	Pralines Gift Box	Germany	2.62	17/10/2022	$209.16	198	$518.013	$168.12
ORD-042648	Yuki Sato	70% Dark Bar	India	3.07	26/03/2022	$82.9	105	$321.7923	$649.05
ORD-042674	Yuki Sato	70% Dark Bar	Japan	14.20	29/10/2022	$59.83	27	$383.0818	$481.71
ORD-042690	Yuki Sato	Milk Classic Bar	India	3.01	01/07/2022	$105.47	80	$240.4171	$233.31
ORD-042705	Yuki Sato	70% Dark Bar	Germany	3.13	31/12/2023	$168.96	241	$752.3537	$993.32
ORD-042790	Yuki Sato	70% Dark Bar	Australia	4.22	10/12/2022	$63.81	124	$523.0759	$1,038.16
ORD-042808	Yuki Sato	Truffle Gift Box	Australia	4.38	20/03/2023	$33.93	62	$271.3699	$742.71
ORD-042833	Yuki Sato	Mixed Assortment Box	Brazil	3.29	30/03/2022	$109.27	333	$1,094.1786	$521.12
ORD-042858	Yuki Sato	70% Dark Bar	Australia	3.36	19/08/2023	$60.14	77	$258.6113	$148.06
ORD-042865	Yuki Sato	70% Dark Bar	Brazil	2.73	20/02/2022	$64.07	149	$406.6398	$488.04
ORD-042886	Yuki Sato	Choco Coins Bag	Brazil	18.58	07/07/2023	$147.26	456	$8,461.1269	$602.32
Total				8,278.98		$130,595.1871	193017	$819,288.9411	$738,555.55

SALES PERSON'S PERFORMANCE

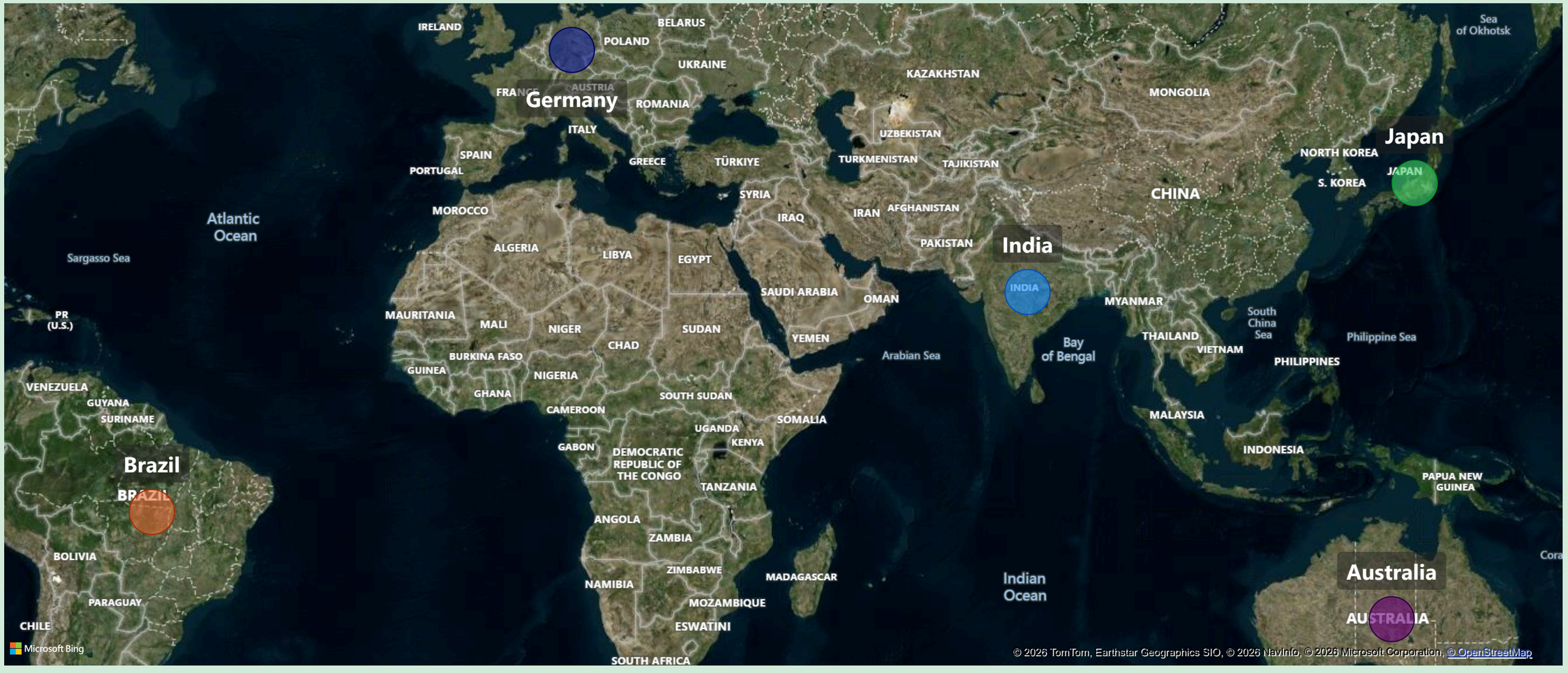
Salesperson	Boxes Shipped	Total Amount
Ananya Gupta	210,040.00	767,510.56
Arjun Mehta	9,435,988.00	34,715,354.85
Beatriz Souza	474,147.00	1,737,665.00
Camila Santos	532,787.00	1,951,011.72
Carlos Mendez	214,626.00	829,637.10
Daniel Weber	283,968.00	993,356.78
Emily Clarke	2,493,747.00	9,137,706.05
Felix Becker	666,702.00	2,477,002.58
Gabriel Silva	284,369.00	1,042,082.56
Hannah Maller	365,330.00	1,366,548.61
Haruto Tanaka	588,130.00	2,152,909.32
Henry Foster	436,021.00	1,586,984.39
Isabella Rossi	254,859.00	995,859.27
James Carter	909,018.00	3,302,537.82
Laura Hoffmann	194,910.00	733,040.74
Lucas Walker	1,735,392.00	6,415,916.19
Lukas Schmidt	399,200.00	1,481,592.03
Michael Johnson	328,875.00	1,209,210.88
Olivia Bennett	331,747.00	1,229,246.37
Priya Sharma	4,101,058.00	14,965,369.38
Rafael Oliveira	764,786.00	2,806,552.97
Rohan Patel	1,355,632.00	5,020,918.47
Sophie Dubois	218,470.00	764,983.71
Sora Watanabe	244,825.00	870,240.62
Yuki Sato	1,092,194.00	4,019,156.97

Top Moving Product

Product	Discount allowed	Marketing Spend	Boxes Shipped	Total Amount
Hazelnut Milk Bar	700.90	508,563.25	596,328.00	2,740,810.38
Salted Caramel Bar	763.04	552,155.28	750,835.00	2,979,137.12
Choco Coins Bag	854.62	618,060.75	1,289,125.00	3,344,853.27
White Choco Bar	930.18	697,456.71	879,918.00	3,685,765.17
85% Dark Bar	1,054.76	761,363.84	802,584.00	4,131,865.97
Almond Crunch Bar	1,204.28	875,594.11	1,584,596.00	4,734,794.59
Mint Dark Bar	1,379.79	1,002,019.62	956,504.00	5,426,142.84
Pralines Gift Box	1,669.64	1,224,458.04	1,580,872.00	6,516,061.71
Milk Classic Bar	2,128.65	1,530,068.47	1,696,890.00	8,287,484.14
Mixed Assortment Box	2,819.82	2,069,368.32	5,334,489.00	10,976,945.20
Truffle Gift Box	4,226.69	3,075,951.52	5,267,480.00	16,587,400.03
70% Dark Bar	8,482.56	6,174,614.13	7,177,200.00	33,161,134.52

Country and Total Amount

Total Amount ● 6063738.129999981 ● 8960869.40999999 ● 12961322.289... ● 27220191.9... ● 47366273....



Measures Calculated

Quantity Measures

27.92M	139.58	3.81K	-1.64K
Total Quantity	Quantity Average	Max Quantity	Min Quantity

Sales Measures

102.57M	28.81	12.14K	112.83M	512.86
Total Sales	MIN Sales	MAX Sales	Sales Target	Average Sales

Discount Measures

26.21K	0.13	0.38	0.00
Total Discount	Average Discount	max Discount	Min Discount

Marketing Spend Measures

19.09M	95.45	798.22	4.30
Total marketing Spend	Average Marketing Spend	Max Marketing Spend	Min Marketing Spend

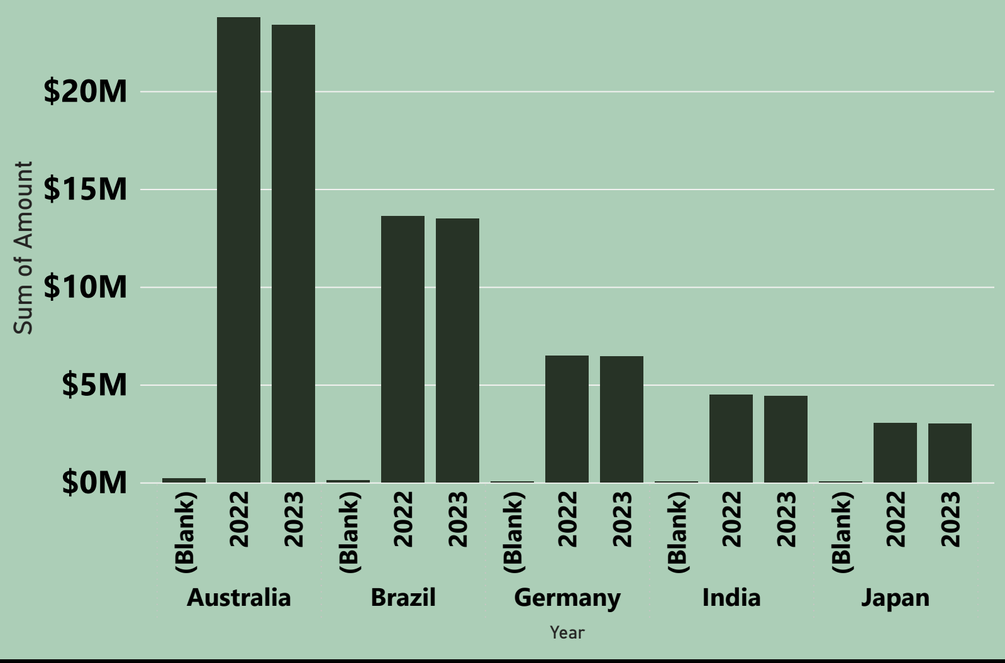
Achievements % , Variance & Total Estimated Sales

90.91	-10.26M	117.98M
Achievements %	Variance	Total of Estimated Sales

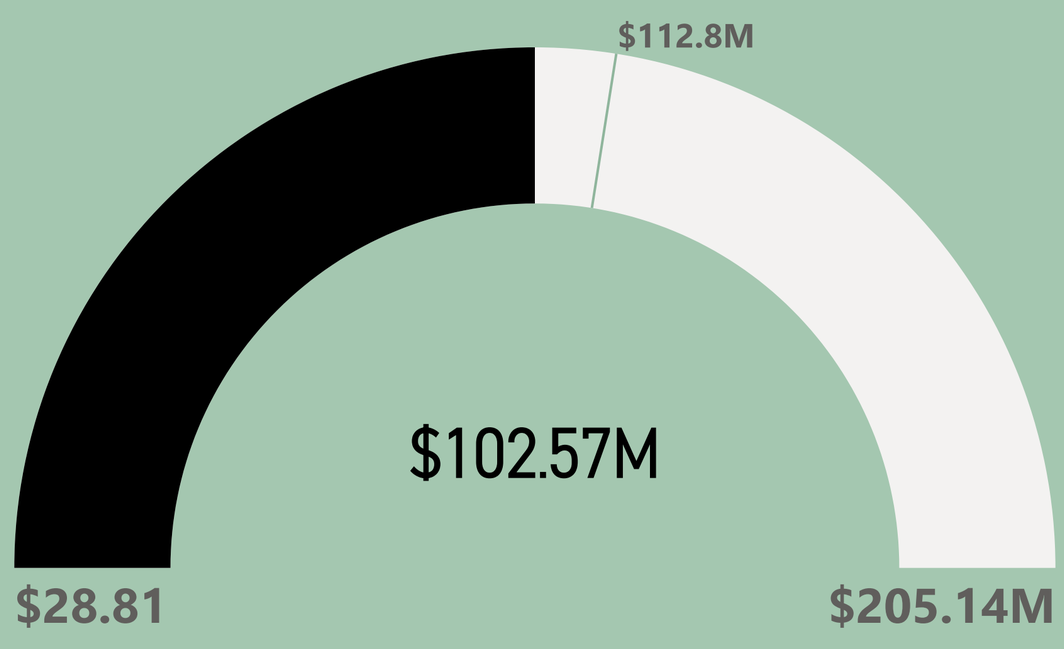
MINI PROJECT DASH BOARD

Chocolate Sales

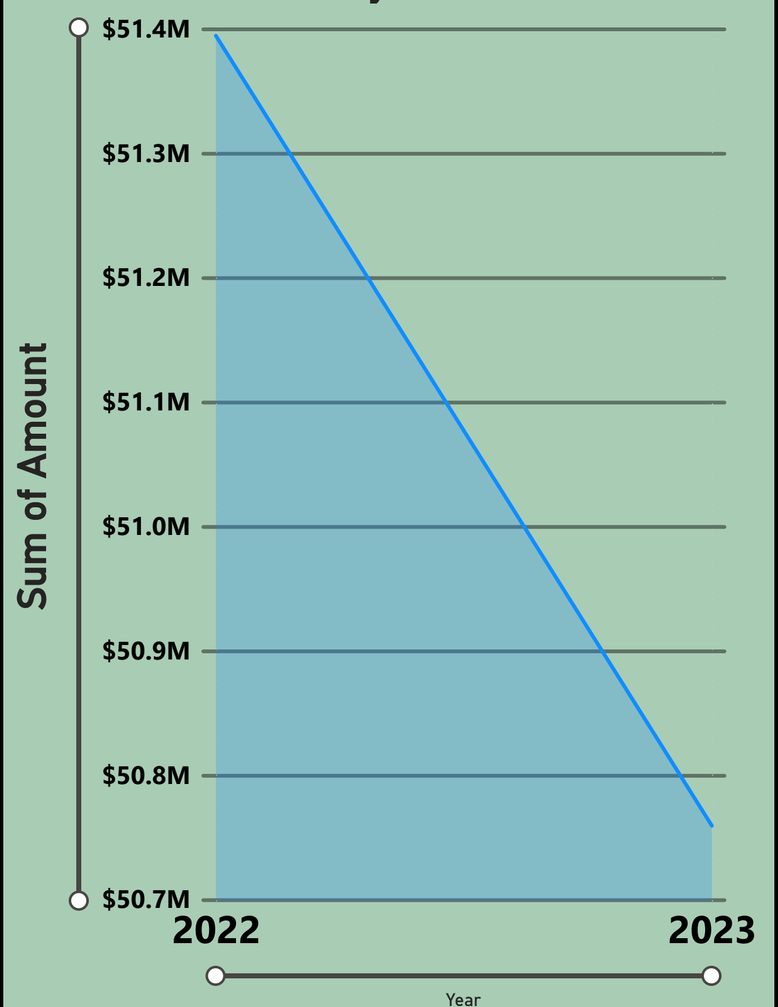
Sum of Amount by Country and Year



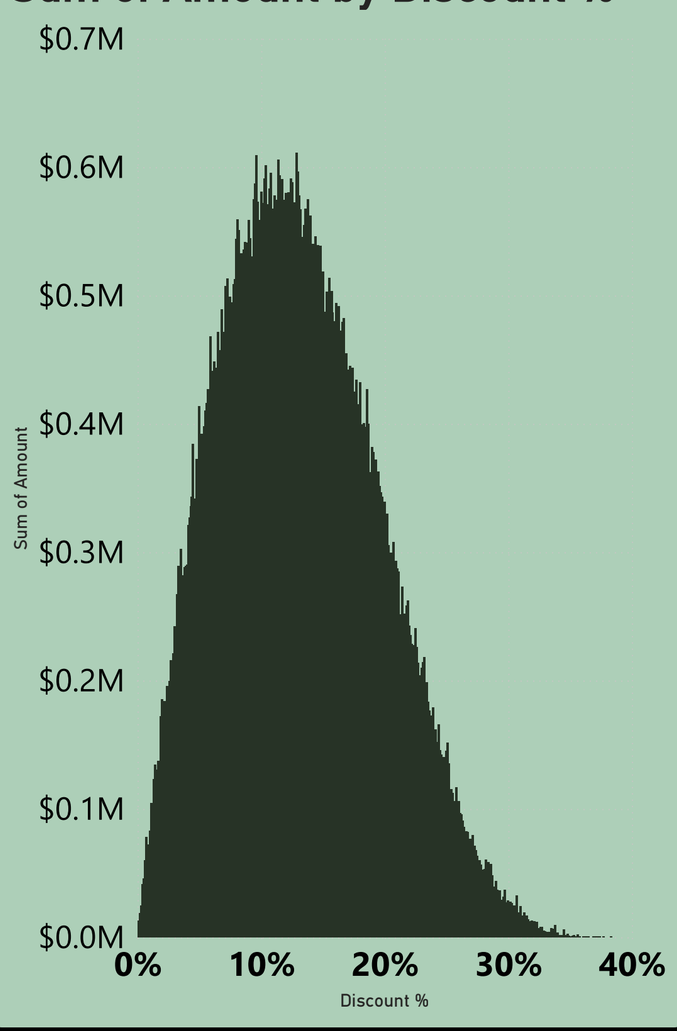
Sum of Amount, MIN Sales and Sales Target



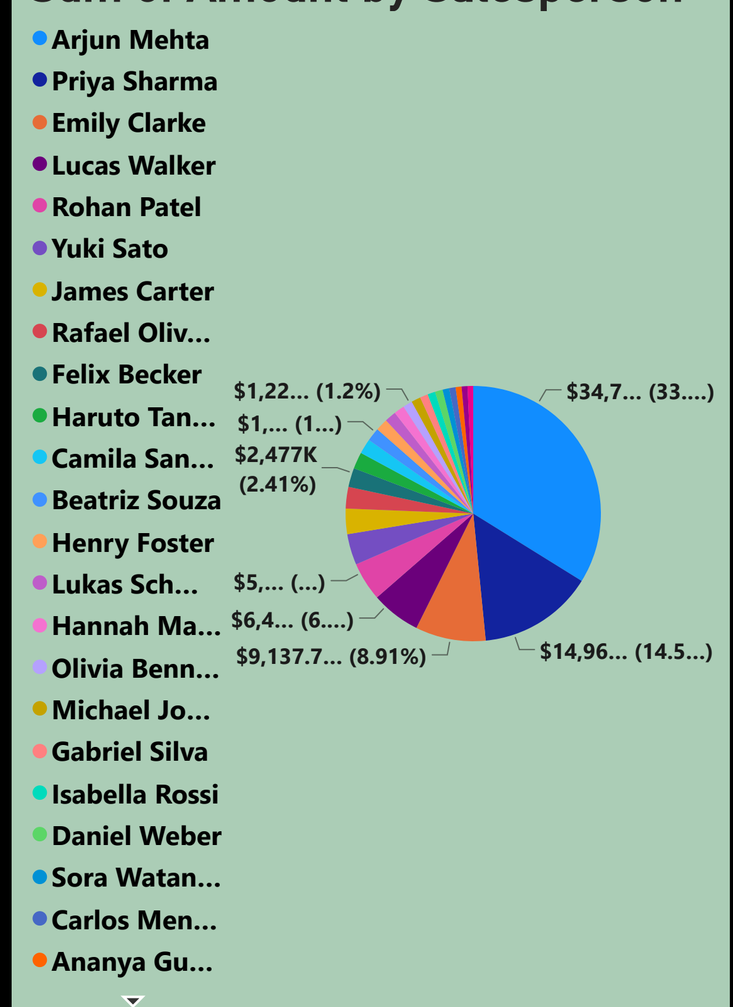
Sum of Amount by Year



Sum of Amount by Discount %



Sum of Amount by Salesperson



MATRIX TABLE

Mini Project Chocolate Sales

Choose year

- ○ --
- ◉ 2022
- ○ 2023

Product	Sum of Amount	Sum of Boxes Shipped	Sum of Discount %	Sum of Marketing Spend
70% Dark Bar	$16,529,814.94	3,496,516.00	423026.88%	$3,084,226.2008
85% Dark Bar	$2,064,140.86	399,274.00	54072.52%	$388,287.7188
Almond Crunch Bar	$2,394,052.83	771,104.00	60552.40%	$439,324.819
Choco Coins Bag	$1,702,640.3	631,100.00	43012.57%	$313,629.1097
Hazelnut Milk Bar	$1,381,457.76	293,622.00	35521.31%	$258,846.6908
Milk Classic Bar	$4,188,185.38	836,467.00	108213.83%	$768,411.8795
Mint Dark Bar	$2,720,579.43	461,137.00	68803.89%	$499,921.3144
Mixed Assortment Box	$5,479,806.23	2,585,614.00	140778.28%	$1,039,570.3284
Pralines Gift Box	$3,253,284.04	774,253.00	83619.73%	$618,267.0038
Salted Caramel Bar	$1,488,285.4	355,461.00	37388.35%	$268,875.8312
Truffle Gift Box	$8,337,767.17	2,556,360.00	210348.28%	$1,525,447.1195
White Choco Bar	$1,854,352.8	419,532.00	46206.63%	$348,099.3744
Total	$51,394,367.14	13,580,440.00	1311544.68%	$9,552,907.3903

DOCUMENTATION

MINI PROJECT CHOCOCLATE SALES

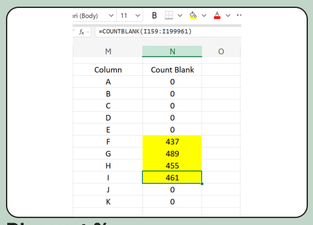
Project Overview

This project analyses Chocolate sales performance using Excel and Power BI.
The objective is to identify sales trends, top-performing products, customer insights, and key business metrics through an interactive dashboard.

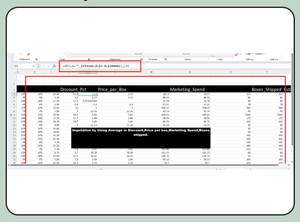
Step 1 - Data Cleaning (Excel)

Removed Duplicates Records
Formatted name column with Find and Replace
Handled Missing values by imputing average
Formatted Discount Column
Created Estimated Net Sales Column
Applied Conditional Formatting
provided VLOOKUP Sample data for further studies

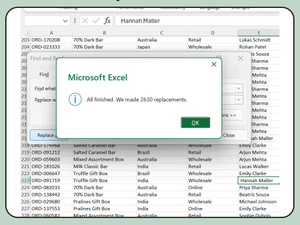
Count Blank



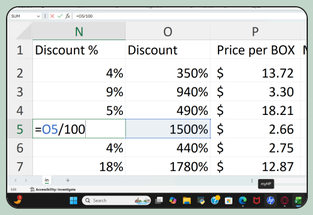
Done Imputation



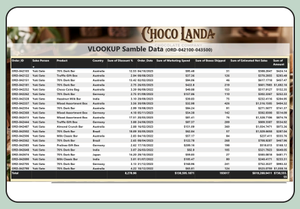
Find and Replace



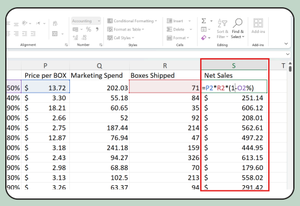
Discount %



VLOOKUP

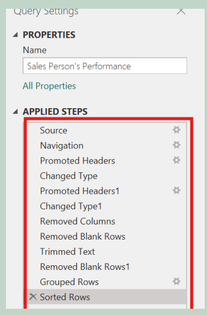


Estimated Net Sales



step 2 - Power Query Data Preparation

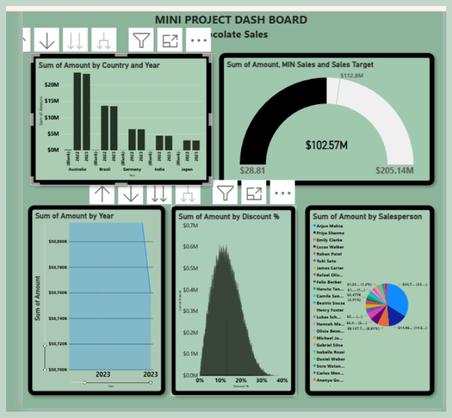
Imported Data from Excel
Cleaned and transformed by correcting data types
Renamed ,merged and formatted columns
Filtered ,sorted and reshaped data for analysis



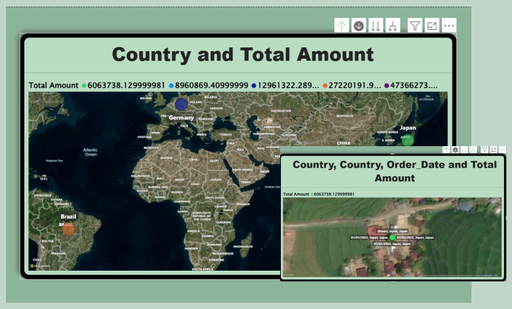
step 3 - Power BI Data Model and Visualisation

Created Calculated Columns and Measures
Created Sales Target by using DAX formula (Total sales*1.1)
Build relationship between tables and developed the data model
Designed interactive Dashboard and reports
Created visuals including charts ,Matrix table, cards, slicers and map
Developed location-wise sales maps for geographic analysis
Created hierarchies and drill through features
Presented business insights using KPI cards and visual analysis

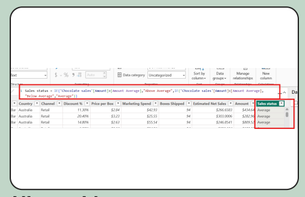
Dash Board with Drill Down Option



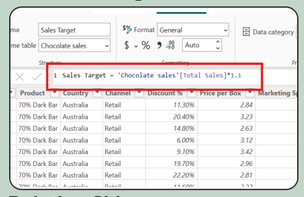
Map drill down



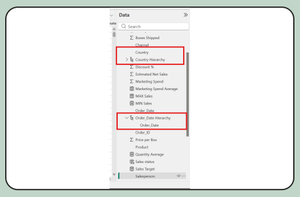
Calculated Column



Set Sales Target



Hierarchies



Relation Ship

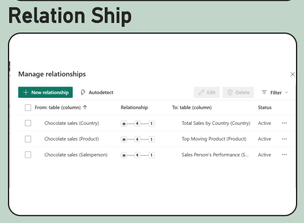
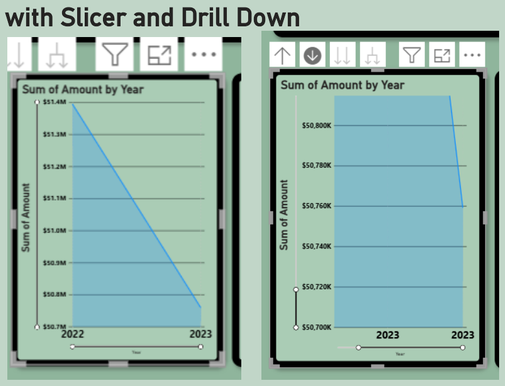
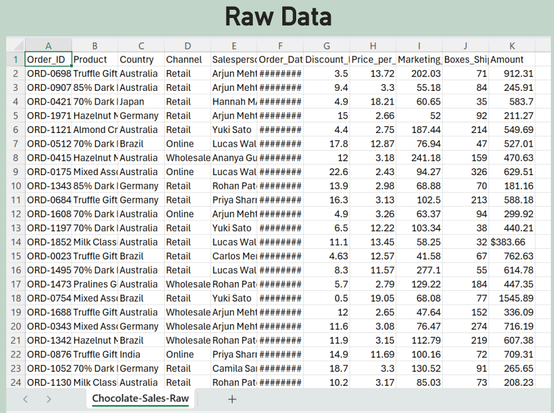


Chart with Slicer and Drill Down

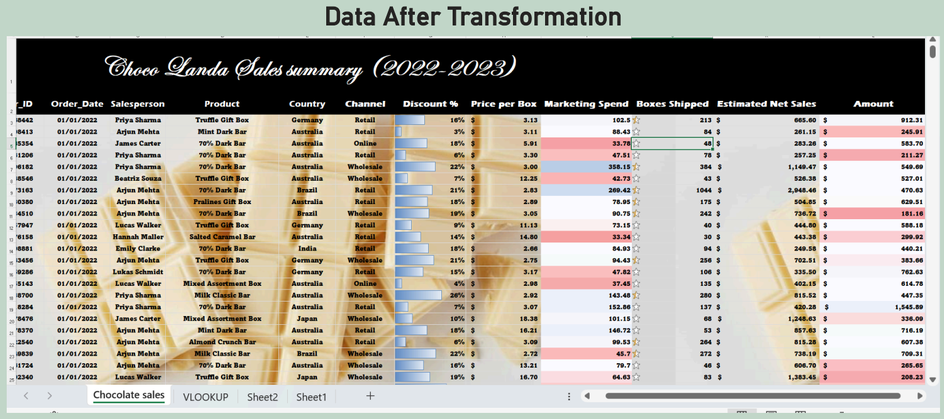


Data Source : https://www.kaggle.com/datasets/arjunmehta1992/chocolate-sales-in-20222023
(200002 Rows)

Raw Data



Data After Transformation



CONCLUSION AND BUSINESS INSIGHT

The analysis suggests that the company's declining performance may be influenced by several factors. High discount rates may have reduced profit margins, while maintaining a larger sales team than necessary could have increase.

If the company is also investing heavily in marketing, it should evaluate whether that spending is generating sufficient returns. Reviewing discount policies, sales staffing, and marketing expenses could help improve profitability and overall business performance.

Also, the analysis shows a significant gap between the estimated net sales and the actual net sales, with actual net sales being considerably lower than expected. This suggests That
the deductions, discounts, returns, or other adjustments may be higher than anticipated. The company should investigate the reasons for these deductions to ensure they are accurate.